

Guidance strategies for controlled Earth reentry of small spacecraft in low Earth orbit

Takumi Saito^{a,*}, Toshinori Kuwahara^{a,**}, Yuji Saito^b, Yuji Sato^c

^a Department of Aerospace Engineering, Tohoku University, Japan

^b The Frontier Research Institute for Interdisciplinary Sciences, Tohoku University, Japan

^c ElevationSpace Inc., Japan

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ABSTRACT

A small spacecraft equipped with a hybrid thruster system providing 100 N-class thrust and a small reentry capsule with a capability of aeroassist maneuver, can deorbit from low Earth orbit, enter a reentry trajectory into Earth atmosphere, and descend toward the target landing point. During the deorbit maneuver, an attitude control system stabilizes the thrust direction, while spin-stabilization method is achieved using cold-gas thrusters. Numerical studies confirm the feasibility of deorbit control, regardless of uncertainties, such as the position of the center of gravity of the spacecraft and the tilt angle of the thrust vector. After a deorbit maneuver, an aeroassist maneuver in the atmosphere can guide a small reentry capsule to a target ground point. In this study, the performance of a real-time predictor-corrector guidance method is improved, a strategy that exhibits high potential as a non-scheduling guidance method in terms of the control accuracy to the target point and the maximum aerodynamic load using a lift-and-drag estimator and terminal bank angle guidance. The resulting algorithm, a robust aeroassist method, outperforms conventional aeroassist maneuvers, even with variations in aerodynamic characteristics due to capsule recession and an offset in the center of gravity of the capsule. The proposed autonomous guidance technique integrates the deorbit and aeroassist maneuver strategies for controlled Earth reentry of small spacecraft with immense potential for application in astronautics.

1. Introduction

Small spacecraft are used in many fields for various applications, including Earth observation in low Earth orbit (LEO), lunar exploration, and deep-space exploration, enabling rapid development at a low cost, and thus play a pivotal role in the space industry. Unlike large spacecraft, stringent constraints limit the inclusion of onboard components in small spacecraft, making orbit insertion relatively complex. A recent publication by the University of Tokyo reports on a three-axis attitude determination and control system using reaction wheels (RWs) [1]. Furthermore, several research groups are exploring the development and application of spacecraft systems comprising a combination of flexible structures, such as membrane films and metal tapes. Besides their application in solar sail systems, the use of these structures as solar panels or in advanced deployment mechanisms is being extensively

investigated [2,3]. Several institutes are investigating orbit control using solar sails, which is particularly useful for small spacecraft with limited actuator capability, particularly where mounting propulsion systems is difficult [4,5]. Additionally, the 6U CubeSat EQUULEUS, developed for deep space exploration, exhibits orbit control using non-chemical propulsion systems in space [6].

A dozen 50 kg-class microsatellites, primarily for Earth observation, have been developed by Tohoku University [7–9]. Among them, RISE-SAT, launched in 2019, enables high-resolution multi-spectral observations and high-accuracy ground pointing control using RWs [10]. The 1U CubeSat FREEDOM and microsatellite ALE-1 enable orbital insertion of membrane deployable structures [11–13]; in these missions, the drag force from atmospheric particles is used to adjust satellite orbits to the desired trajectory [14,15]. Furthermore, the microsatellite ALE-2 uses 3 mN-class cold-gas thrusters for orbit control. Over the past three years,

Abbreviations: LEO, Low Earth Orbit; RW, Reaction Wheel; HTV, H-II Transfer Vehicle; HTS, Hybrid Thruster System; LVLH, Local Vertical Local Horizon; RCS, Reaction Control System.

* Corresponding author. Aoba 6-6-11, Aramaki, Aoba-ku, Sendai, 980-8579, Japan.

** Corresponding author

E-mail addresses: saito.takumi.t7@dc.tohoku.ac.jp (T. Saito), toshinori.kuwahara.b3@tohoku.ac.jp (T. Kuwahara).

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thruster operations have been used for the orbital ascent and attitude control of spacecraft more than 100 times [16]. Additionally, studies are focused on miniaturizing this propulsion system to the 1U-class and designing an orbit control system for a 3U CubeSat for the development of an asynchronous one-way ranging measurement method [17,18].

Recent studies reported orbit control systems for small spacecraft, such as the microsatellite with air-launchable re-entry capabilities (MISTRAL), where the ballistic coefficient is controlled through structural deformation [19,20]. The release-to-ground-landing of this spacecraft, which passively enters the reentry trajectory on release from the International Space Station, is a long-term process. A small reentry capsule (HSRC) integrated into the H-II transfer vehicle (HTV), a 200 kg-class entry capsule developed by the Japan Aerospace Exploration Agency (JAXA), exhibits aeroassist maneuvers [21–23]. The HSRC cannot directly insert itself into the reentry trajectory and instead relies on multiple deorbit maneuvers by the HTV. Controlled Earth reentry typically requires a significant reduction in a spacecraft's kinetic energy, a high-thrust propulsion system, and large amounts of propellant. However, this approach is unsuitable for small spacecraft, making it critical to address this challenge, which has not yet been reported in the literature.

Recent studies further aimed to develop a high-thrust propulsion system, known as a hybrid thruster system (HTS), that can enable small spacecraft to exhibit an active deorbit maneuver. Unlike conventional chemical engines, the hybrid thruster system uses plastic fuel; moreover, this type of rocket engine can be miniaturized for small spacecraft [24–27]. The hybrid thruster system, designed as an actuator for deorbit maneuvers capable of generating a 100 N-class thrust, was originally developed for the Earth reentry mission of a 220 kg spacecraft that can deorbit with a 140 m/s velocity increment from an altitude of 450 km from LEO. A 30 kg entry capsule can fly with zero-lift guidance. Zero-lift guidance is a simple aeroassist method in which the capsule utilizes spin stabilization for attitude control instead of manipulating aerodynamic forces. Although this approach minimizes landing dispersion as a constant angle of attack is maintained, this ballistic flight limits both flight distance and landing accuracy due to non-negligible errors in atmospheric density and the aerodynamic characteristics of the capsule. Under zero-lift guidance, the landing dispersion may exceed 100 km from the target point, making the ground retrieval of the vehicle challenging. For example, the landing ellipse of the Hayabusa2 capsule, which uses this method of atmospheric entry, can exceed 150 km [28].

Two approaches have been developed for controlled reentry guidance for aeroassist maneuvers: an explicit method using a reference trajectory and an implicit method using numerical integration, which does not follow a fixed trajectory. The explicit method, which reduces onboard computational costs considering the flight profile in terms of flight range, altitude rate, and drag acceleration, has been employed since the Apollo mission era for Earth reentry and Mars entry [29–32]. In contrast, implicit guidance requires several onboard calculations, such as those on variable gains for the flight range or predictions of the vehicle trajectory. The Orion capsule applies implicit guidance using the numerical predictor-corrector method for skipping reentry guidance [33–35]. The HSRC project proposes a real-time numerical integration guidance that predicts the capsule trajectory. The in-orbit feasibility of this guidance, which entails a relatively high computational cost for the onboard computer, has been confirmed.

This paper discusses the feasibility of the controlled Earth reentry of a small spacecraft; specifically, a 300 kg spacecraft that includes a 150 kg entry capsule. Chapter 2 describes the deorbit maneuver and the effect of attitude motion on the reentry trajectory during this process. Chapter 3 outlines the two aeroassist guidance algorithms and provides an overview of aeroassist maneuvers. Chapter 4 evaluates the guidance performance of both the explicit and real-time predictor-corrector methods in terms of robustness against errors in atmospheric density and the aerodynamic characteristics of the vehicle. Finally, a maneuver strategy that integrates both deorbit and aeroassist maneuvers is described.

2. Overview of small spacecraft deorbit maneuvers

2.1. Small spacecraft system specifications

Fig. 1 shows the small spacecraft system used in this study; the system specifications are summarized in Table 1, presented in the body coordinate frame of the spacecraft, where the X_B -axis is opposite to the entry capsule. The system has the total mass of 330 kg, including a 150 kg entry capsule. Two types of thrusters are mounted on the vehicle for the deorbit maneuver: a hybrid thruster system that functions as the main actuator and ensures an adequate velocity increment, and a cold-gas thruster system that functions as a supplementary actuator for attitude control. Here, for simplicity, the hybrid thruster system and cold-gas thrusters are labeled HTS and RCS, respectively. Only one module of the hybrid thruster system, which is expected to weigh approximately 100 kg, is implemented in the vehicle to facilitate a single hybrid thruster firing operation. Eight cold-gas thrusters are installed for the spin control of the body axis: RCS 1–4 are used for roll spin, while RCS 5–8 are used for pitch and yaw de-spin. In addition, six units are mounted on the capsule model for attitude control. The hybrid thruster is aligned along the X_B -axis in the body coordinate frame, and the entry capsule is assumed to be of the same scale as the HSRC system and positioned opposite the hybrid thruster, so it remains separated from the vehicle. In this study, the drag coefficient of the satellite was defined based on the flat plate assumption in the atmosphere, and the capsule value was based on the characteristics of the HSRC. The characteristics of the HTS and RCS are listed in Table 2.

2.2. Deorbit maneuver strategy

This study considers two factors that contribute to the attitude disturbance moment owing to the HTS: the offset of the center of gravity and the elevation angle of the HTS thrust direction from the X_B -axis. Fig. 2 depicts the following concepts: the disturbance moment generated due to the offset of the center of gravity and the thrust vector of the HTS deviating from the X_B -axis due to the elevation angle. As shown in Fig. 2(a), when the initial position of the center of gravity, G_1 , is shifted to G_2 by a distance l in the $Y_B Z_B$ plane and a distance h along the X_B -axis, a moment occurs owing to the HTS thrust vector. Fig. 2(b) shows that a moment is generated when the initial thrust vector, defined as $F_{th,\tau=0}$, deviates to $F_{th,\tau}$, regardless of an offset in the center of gravity.

These factors are inevitable in hardware development; therefore, attitude control is essential during the HTS firing. The magnitude of these disturbance moments is significantly larger than the angular momentum, which can be counteracted by the RWs mounted on a small spacecraft. An attitude control approach using RCS was used for the spin-up maneuver along the X_B -axis, which was aligned with the HTS thrust vector. After completing the HTS firing, the angular velocity of the spacecraft along each axis must be significantly reduced to facilitate

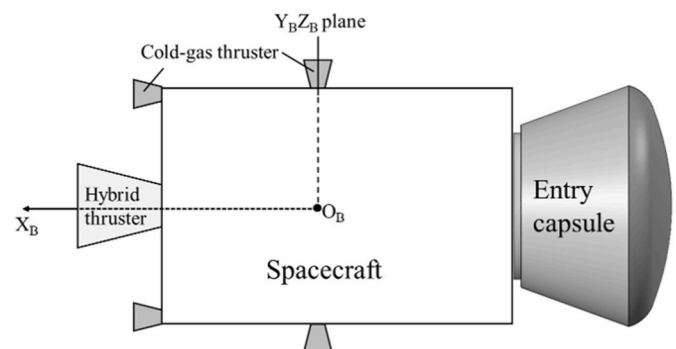


Fig. 1. Structural design of small spacecraft with hybrid thruster system (HTS) and small capsule.

Table 1
Specifications of small spacecraft used for analysis.

Property	Satellite	Capsule
Mass [kg]	180	150
Moment of inertia [kgm ²]	$\begin{bmatrix} 23.3 & 0.119 & -0.127 \\ 0.119 & 45.5 & 0.00930 \\ -0.127 & 0.00930 & 49.7 \end{bmatrix}$	$\begin{bmatrix} 2.85 & -0.0115 & 0.00580 \\ -0.0115 & 2.92 & 0.00580 \\ 0.00580 & 0.00580 & 2.94 \end{bmatrix}$
Size [m]	Width: 0.800 Length: 1.25 Height: 0.845	Diameter: 0.840 Height: 0.657
Reference area [m ²]	0.640	0.554
Drag coefficient [–]	2.2	1.3

Table 2
Characteristics of hybrid thruster and cold-gas thruster systems.

Property	Hybrid thruster	Cold-gas thruster on satellite	Cold-gas thruster on capsule
Thrust [N]	172	8.4	2.4
Position [m]	$[0.474 \ 0 \ 0]$	$\begin{bmatrix} \text{TH1} \\ \text{TH2} \\ \text{TH3} \\ \text{TH4} \\ \text{TH5} \\ \text{TH6} \\ \text{TH7} \\ \text{TH8} \end{bmatrix} = \begin{bmatrix} 0.017 & 0.436 & 0.371 \\ 0.017 & 0.436 & -0.371 \\ 0.017 & -0.436 & 0.371 \\ 0.017 & -0.436 & -0.371 \\ 0.553 & 0.372 & 0.372 \\ 0.553 & 0.372 & -0.372 \\ 0.553 & -0.372 & 0.372 \\ 0.553 & -0.372 & -0.372 \end{bmatrix}$	$\begin{bmatrix} \text{TH1} \\ \text{TH2} \\ \text{TH3} \\ \text{TH4} \\ \text{TH5} \\ \text{TH6} \end{bmatrix} = \begin{bmatrix} 0 & -0.300 & -0.300 \\ 0 & 0.300 & -0.300 \\ 0 & 0 & 0.300 \\ 0 & 0.300 & 0 \\ 0 & 0 & -0.300 \\ 0 & -0.300 & 0 \end{bmatrix}$
Residual thrust [N]	0.0100	–	–
Mass flow rate [g/s]	–	14.5	3.07

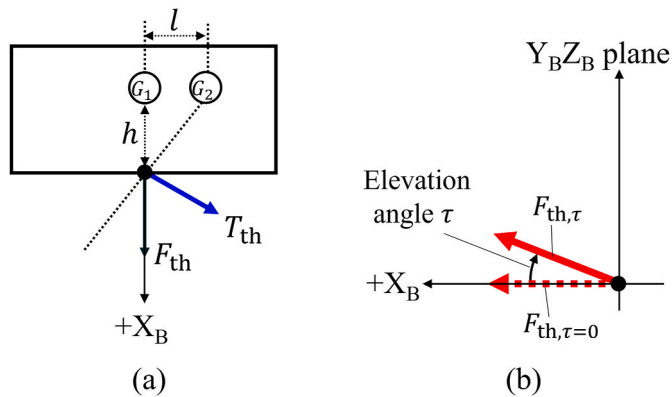


Fig. 2. Disturbance moment generated by HTS during deorbit maneuver: (a) Moment caused by an offset of the center of gravity; (b) Moment caused by the tilt angle of the thrust vector.

a capsule releasing maneuver using RWs, which necessitates a de-spin maneuver.

The deorbit maneuver was assumed to take place in a sun-synchronous orbit at the altitude of 450 km. The spacecraft transitions from its initial orbit to an Earth reentry orbit through a deorbit

sequence. Fig. 3 illustrates the sequence of this deorbit maneuver.

- Phase 1 involves an attitude maneuver in which the spacecraft tracks the target attitude within a local vertical local horizontal (LVLH) frame using RWs.
- Phase 2 is the spin-up maneuver along the X_B -axis for the attitude of the spacecraft; it maintains the inertia fixed attitude and aligns the X_B -axis with the along-track direction of the LVLH frame.
- Phase 3 is an HTS-based deorbit maneuver. The firing duration is assumed to be 210 s based on development specifications.

After the main firing of HTS in Phase 3, due to solid fuel and ablator gasification, a residual thrust of a few mN is expected to be generated.

- Phase 4 is the standby period for the residual thrust to dissipate over 540 s.
- Phase 5 involves an RCS-based de-spin maneuver along the three axes of the body frame.
- In Phase 6, the desired attitude condition for capsule release is established through an attitude maneuver.

2.3. Numerical analysis of deorbit maneuver

The performance of the deorbit maneuver was evaluated using a 6

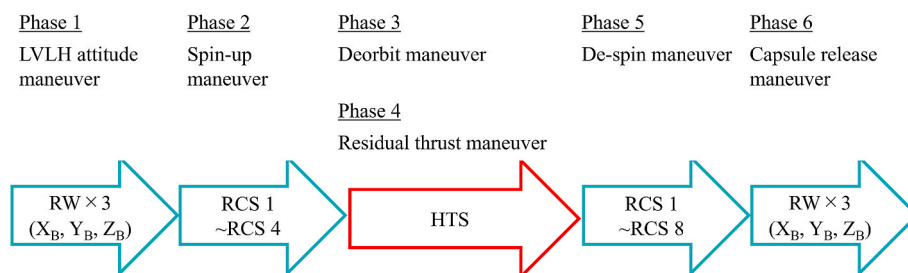


Fig. 3. Deorbit maneuver sequence.

degree-of-freedom simulation. First, the attitude dynamics of a rigid-body spacecraft can be described by the following nonlinear differential equation:

$$\mathbf{J}_{\text{sat}} \dot{\boldsymbol{\omega}} = \mathbf{T}_{\text{HTS}} + \mathbf{T}_{\text{RCS}} - \boldsymbol{\Omega}_{\omega}^{\times} (\mathbf{J}_{\text{sat}} \boldsymbol{\omega} + \mathbf{H}_{\text{RW}}), \quad (1)$$

where $\mathbf{J}_{\text{sat}}$ is the moment of inertia of the spacecraft (as listed in Table 1); $\boldsymbol{\omega} \in \mathbb{R}^{3 \times 1}$ is the angular velocity of the spacecraft in the body frame; $\boldsymbol{\Omega}_{\omega}^{\times}$ is the skew-symmetric matrix composed of the angular velocity about the X_B -, Y_B -, and Z_B -axes, as expressed in Equation (2); $\mathbf{T}_{\text{HTS}} \in \mathbb{R}^{3 \times 1}$ is the torque generated by the HTS, as described in Section 2.2; $\mathbf{T}_{\text{RCS}} \in \mathbb{R}^{3 \times 1}$ is the torque generated by the RCSs; and $\mathbf{H}_{\text{RW}} \in \mathbb{R}^{3 \times 1}$ is the angular momentum of the RWs.

$$\boldsymbol{\Omega}_{\omega}^{\times} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad (2)$$

The nonlinear differential equation for the attitude kinematics of the spacecraft is expressed in Equation (3):

$$\dot{\mathbf{q}} = \frac{1}{2} \boldsymbol{\Omega}_{\mathbf{q}}^{\times} \mathbf{q}, \quad (3)$$

where $\mathbf{q} \in \mathbb{R}^{4 \times 1}$ is the unit quaternion and $\boldsymbol{\Omega}_{\mathbf{q}}^{\times}$ denotes the skew-symmetric matrix, which is given by

$$\boldsymbol{\Omega}_{\mathbf{q}}^{\times} = \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \quad (4)$$

The control algorithm for the RCSs is based on the quaternion feedback regulator of Weiss, expressed as:

$$\begin{aligned} \mathbf{q}_E &= \mathbf{q}_B^{-1} \otimes \mathbf{q}_D, \\ \boldsymbol{\omega}_E &= \boldsymbol{\omega}_D - \boldsymbol{\omega}_B, \end{aligned} \quad (5)$$

where the subscripts E and D represent the error and desired values, respectively, and $\otimes$ denotes quaternion multiplication. The PID controller is applied to calculate the RCS torque.

The spin axis is assumed to be along the X_B -axis of the spacecraft. The offset of the spin axis is evaluated as the nutation angle, δ , using the moments of inertia of the spacecraft about the X_B -, Y_B -, and Z_B -axes, written:

$$\delta = \tan^{-1} \left(\frac{\left\{ (J_y + J_z)/2 \right\} \sqrt{\omega_y^2 + \omega_z^2}}{J_x \omega_x} \right) \quad (6)$$

The orbital dynamics equation of motion for the spacecraft is expressed as:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r} + \mathbf{a}_{\text{HTS}} + \mathbf{a}_{\text{dis}}, \quad (7)$$

where μ is the gravitational parameter based on EGM-96, $\mathbf{r}$ is the position vector of the spacecraft in the ECI coordinate frame, $\mathbf{a}_{\text{HTS}}$ is the thrust acceleration from the HTS, as represented in Equation (8).

$$\mathbf{a}_{\text{HTS}} = \begin{bmatrix} \cos \tau \\ \sin \tau \\ 0 \end{bmatrix} \frac{F_{\text{HTS}}}{m}, \quad (8)$$

where τ is the elevation angle described in Fig. 2(b), F_{HTS} is the thrust

Table 3
Deorbit maneuver conditions.

Property	Values
Initial epoch	2026-07-05, 22:35:00, UTC
Position in ECI frame[km]	[−672.822 −4,497.903 −5,080.118]
Velocity in ECI frame [km/s]	[−2.212 −5.332 5.006]
Deorbit ignition epoch	2026-07-05, 22:39:30, UTC
Center of gravity [mm]	Case 1, 2: [0 0 0], Case 3, 4: [30 20 20]
Elevation angle [°]	Case 1, 3: 0, Case 2, 4: 2.2
Initial quaternion [−]	[0 0 0 1]
Initial angular velocity [m/s]	[0.10 0.10 0.10]

vector of the HTS, and m is the mass of the spacecraft. $\mathbf{a}_{\text{dis}}$ represents the external accelerations due to aerodynamic drag, ellipsoid perturbation, and perturbing forces from the sun and Moon, given by:

$$\mathbf{a}_{\text{dis}} = \mathbf{a}_{\text{drag}} + \mathbf{a}_{\text{potential}} + \mathbf{a}_{\text{sun}} + \mathbf{a}_{\text{moon}} \quad (9)$$

Aerodynamic drag acceleration $\mathbf{a}_{\text{drag}}$ is calculated using atmospheric density modeled according to the U.S. Standard Atmosphere 1976, the drag coefficient described in Table 1, the reference area of the spacecraft, and the velocity vector in the ECI coordinate frame. $\mathbf{a}_{\text{potential}}$ was derived from the derivative of the perturbations using the ellipsoid model, including up to harmonic term J7. $\mathbf{a}_{\text{sun}}$ and $\mathbf{a}_{\text{moon}}$ were derived from the gravitational parameters of the Sun and Moon and the position vectors modeled in the ephemeris model of the Sun and Moon. Second, the deorbit conditions, including the initial orbit parameters of the spacecraft and the firing timing of the deorbit maneuver, were defined (Table 3). Then, to assess the attitude control performance of the spin stabilization, the attitude disturbance conditions during the deorbit maneuver were assumed. The position of the center of gravity was defined as either the origin of the body coordinate frame or the position that is 20 mm offset along the Y_B and Z_B -axes and 30 mm offset along the X_B -axis, representing the maximum offset condition where errors may occur in the management of the center of gravity. In addition, the elevation angle of either 0° or 2.2° was assumed as the maximum error angle. Four specific cases were analyzed in this study: Case 1 represents ideal conditions without any center of gravity or elevation angle offset; Case 2 considers the elevation angle of 2.2° and no offset in the center of gravity; Case 3 considers an offset from the origin in the body frame with no elevation angle offset; and Case 4 represents the worst-case scenario with maximum attitude disturbances, considering a maximum offset and the elevation angle of 2.2°. These analyses were conducted on a laptop computer with a 2.7 GHz Intel Core i7 processor using MATLAB R2022b.

Fig. 4 shows the rotational motion of the spacecraft in Case 1. Fig. 4(a) shows the profile of the angular velocity along the X_B -axis; the black dot line represents the phase profile shown in Fig. 4. Here, the spacecraft can be controlled to maintain the LVLH attitude in Phase 1. In Phase 2, the spin rate along the X_B -axis increases to 180°/s, transitioning the sequence to Phase 3. During the deorbit maneuver, the spin rate (180°/s) and angular velocities in the Y_B and Z_B axes remain constant. In Phase 5, the angular velocities along the three axes decrease through the de-spin maneuver, leading to Phase 6 capsule separation. Fig. 4(b) shows the projection of the trajectory of the X_B -axis onto the $Y_B Z_B$ plane. The spin motion is stable, and this method enables attitude stabilization during the deorbit maneuver even though the offset of the spin axis is toward the X_B -axis, which arises because the moment of inertia of the spacecraft includes off-diagonal elements, as listed in Table 1. Fig. 5(a) shows the velocity increment profile in the along-track, cross-track, and radial directions of the LVLH frame. During Phase 3, the velocity increment in the along-track direction exhibits values up to 110 m/s,

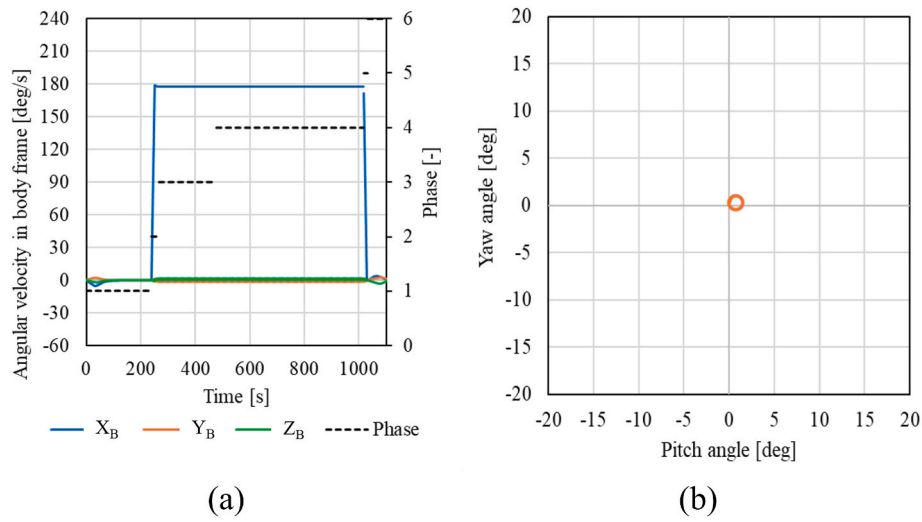


Fig. 4. Rotation motion of spacecraft in Case 1: (a) Angular velocity profile; (b) Projection of spin-axis motion in the $Y_B Z_B$ plane.

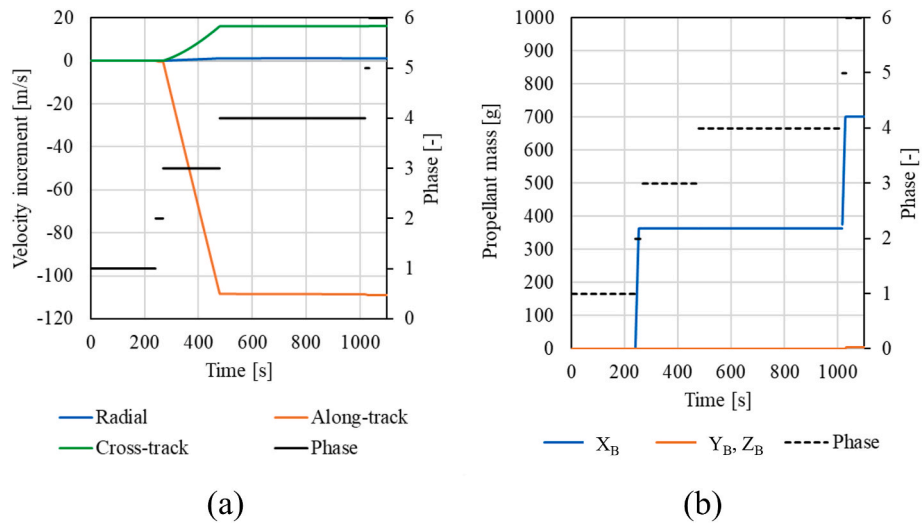


Fig. 5. Simulation results in Case 1: (a) Velocity increment in the LVLH coordinate frame; (b) Propellant mass consumed in RCS.

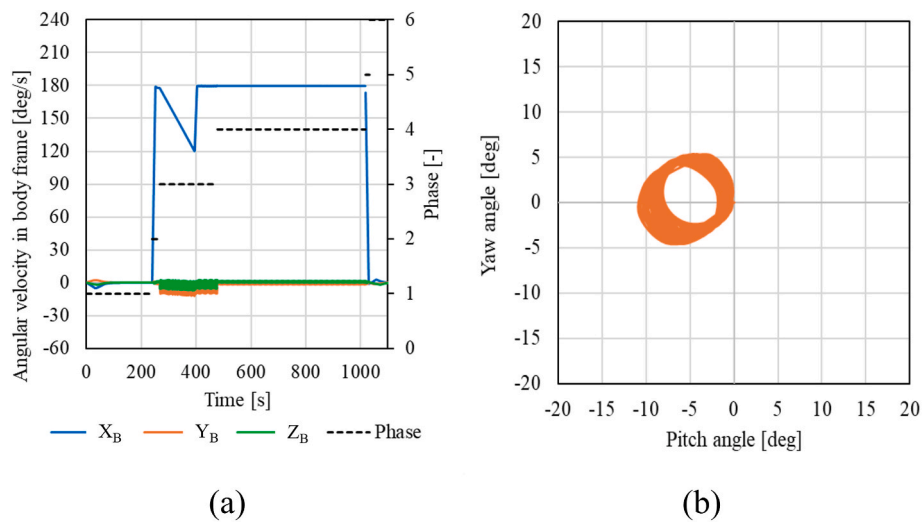


Fig. 6. Rotation motion of spacecraft under the attitude disturbance conditions assumed in Case 4: (a) Angular velocity profile; (b) Projection of the spin-axis motion in the $Y_B Z_B$ plane.

whereas that in the cross-track component is only 15 m/s. Moreover, the radial component is negligible, and the residual thrust does not significantly affect the velocity increment during Phase 4. Fig. 5(b) shows the RCS propellant consumption. Nitrogen gas was assumed to be the RCS propellant. As shown in the figure, most of the propellant is consumed along the X_B -axis; propellant consumption along the Y_B and Z_B axes is minimal.

In both Cases 2 and 3, the spin rate along the X_B -axis decreases following HTS firing owing to the attitude disturbance moment caused by the elevation angle and center of gravity offset. These factors gradually degrade spin stability, potentially leading to attitude instability during the deorbit maneuver.

Fig. 6 shows the rotational behavior of the spacecraft in Case 4, and Fig. 7 shows the velocity increment profile in each direction and the propellant mass consumption. Fig. 6(a) shows that the spin rate decreases rapidly during HTS firing. To prevent attitude instability, a spin-up correction maneuver is implemented when the spin rate becomes $<120^\circ/\text{s}$. Fig. 6(b) shows the spin axis rotation projected onto the $Y_B Z_B$ plane, which indicates a further offset from the origin of the body frame. This rotation occurs through a nutation angle, which is the angle between the spin and X_B -axis. In this study, the minimum spin rate threshold was determined via sensitivity analysis. In Fig. 7(a), the velocity increment in the along-track direction is consistent with that in Case 1, indicating that disturbances, such as the center of gravity offset and thrust-vector error, influence the velocity increment negligibly. Fig. 7(b) shows that ~ 200 g of the propellant mass is consumed during the spin rate correction maneuver. Thus, maintaining an adequate propellant margin can ensure attitude stability, even under the disturbance conditions assumed in Case 4.

2.4. Conclusion of deorbit maneuver

After separating from the satellite module, the capsule reenters the atmosphere of the Earth. Table 4 lists the orbital conditions of the capsule at the entry interface altitude of 120 km for all four cases. The range-to-go distance represents the range between the current position and the target position of the capsule on the ground, which is calculated after assuming a circular distance. The target point is located on the southeastern side of Minamitorishima Island (22.74° N, 158.78° E). Fig. 8 shows the ground track of the spacecraft in Case 4, as well as the target landing site. The velocity increment in the along-track direction varies owing to attitude disturbance factors, causing deviations in the entry trajectory. The maximum dispersion in the range-to-go between

Table 4

Entry interface conditions after deorbit maneuver in Case 1–4.

Parameter	Case 1	Case 2	Case 3	Case 4
Geodetic altitude [km]	121.13	121.06	121.01	121.15
Longitude [$^\circ$]	176.04	176.22	176.26	176.44
Geodetic latitude [$^\circ$]	64.22	64.47	64.54	64.79
Velocity [km/s]	7.97	7.97	7.97	7.97
Flight path angle [$^\circ$]	−1.16	−1.16	−1.16	−1.17
Velocity azimuth angle [$^\circ$]	−161.74	−161.61	−161.59	−161.46
Range-to-go [km]	4,852	4,882	4,890	4,920

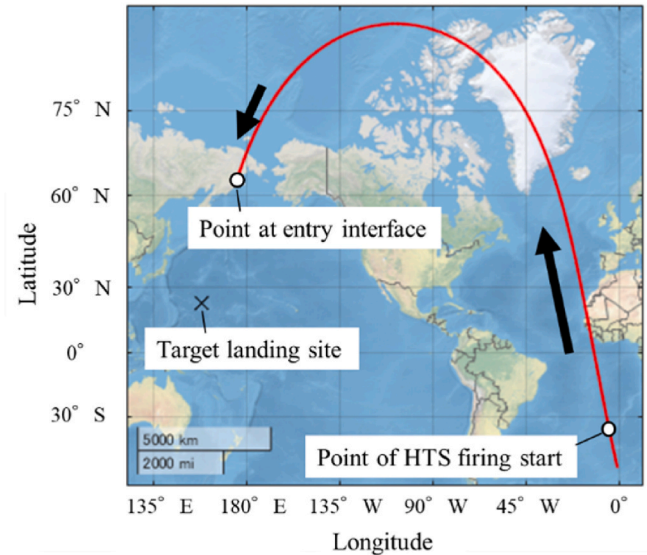


Fig. 8. Ground track of spacecraft in Case 4 and target landing site.

Case 1 and Case 4 is ~ 70 km. Thus, the proposed deorbit maneuvering approach shows controlled accuracy within several tens of kilometers, thereby ensuring robustness with respect to uncertainties in attitude conditions.

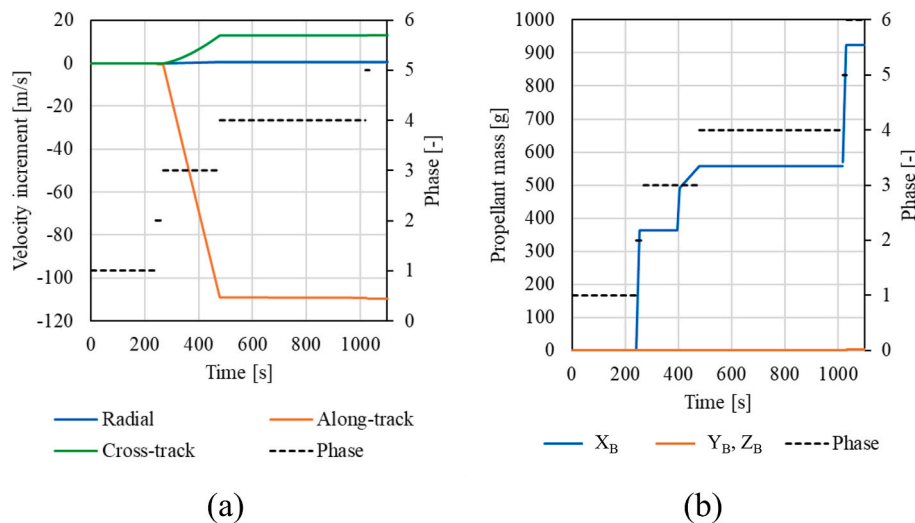


Fig. 7. Simulation results under the attitude disturbance conditions assumed in Case 4: (a) Velocity increment in the LVLH coordinate frame; (b) Propellant mass consumed in RCSs.

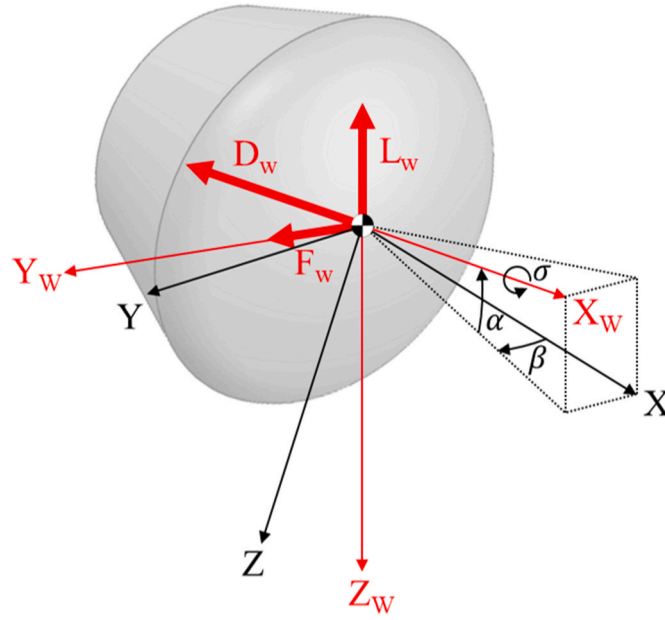


Fig. 9. Definition of the capsule coordinate systems in the wind axis coordinate frame and the body coordinate frame.

3. Guidance technique for aeroassist maneuver

3.1. Equation of motion for entry vehicles

The translational motion of the entry vehicle is described in the wind axis coordinate frame with the subscript W, where the X_W -axis is aligned with the velocity direction, the Y_W -axis corresponds to the side-force direction, and the Z_W -axis opposes with the lift force direction. The onboard measurable lift and drag forces are represented in the body coordinate frame of the vehicle. The wind axis frame is rotated by three angles: bank angle σ , angle of attack α , and sideslip angle β . Equation (10) represents the equation of motion of the vehicle in the atmosphere of the Earth:

$$\begin{aligned} \dot{r} &= v \sin \gamma, \\ \dot{\phi} &= \frac{v \cos \gamma \cos \psi}{r}, \\ \dot{\theta} &= \frac{v \cos \gamma \sin \psi}{r \cos \phi}, \\ m \dot{v} &= -D_W - mg \sin \gamma \\ &\quad - m \omega^2 r \cos \phi (\cos \gamma \cos \psi \sin \phi - \sin \gamma \cos \phi), \\ m v \cos \gamma \dot{\psi} &= F_W + \frac{m v^2}{r} \cos^2 \gamma \sin \psi \tan \phi + m \omega^2 r \sin \psi \sin \phi \cos \phi \\ &\quad - 2 m \omega v (\sin \gamma \cos \psi \cos \phi - \cos \gamma \sin \phi), \\ m v \dot{\gamma} &= L_W - mg \cos \gamma + \frac{m v^2}{r} \cos \gamma + 2 m \omega v \sin \psi \cos \phi \\ &\quad + m \omega^2 r \cos \phi (\sin \gamma \cos \psi \sin \phi + \cos \gamma \cos \phi), \end{aligned} \quad (10)$$

$$\begin{bmatrix} D_W \\ F_W \\ L_W \end{bmatrix} = \begin{bmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ \sin \alpha \sin \alpha \cos \beta - \cos \alpha \sin \beta & \sin \alpha \sin \alpha \sin \beta + \cos \alpha \cos \beta & \sin \alpha \cos \alpha \\ \cos \alpha \sin \alpha \cos \beta + \sin \alpha \sin \beta & \cos \alpha \sin \alpha \sin \beta - \sin \alpha \cos \beta & \cos \alpha \cos \alpha \end{bmatrix} \begin{bmatrix} D \\ F \\ L \end{bmatrix} \quad (12)$$

where r represents the radius from the center of the Earth, v is the velocity magnitude, ϕ is the geodetic latitude, θ is the longitude, γ is the flight path angle, ψ is the velocity azimuth angle, m is the vehicle's mass; g and ω are the gravitational acceleration and angular velocity of the Earth, respectively; and D_W , F_W , and L_W are the drag, side, and lift forces

in the wind axis coordinate frame, respectively. Fig. 9 illustrates the definition of these coordinate systems.

Equation (11) defines the magnitudes of the lift force and drag forces in the body coordinate frame:

$$L = \frac{\rho v^2 C_L S}{2}, \quad D = \frac{\rho v^2 C_D S}{2}, \quad (11)$$

where ρ represents the atmospheric density, S is the reference cross-sectional area of the vehicle, and C_L and C_D are the lift and drag coefficients, respectively.

Equation (12) expresses the force components in the wind axis frame, incorporating the attitude rotation of the vehicle, specifically, the angle of attack, sideslip angle, and bank angle from the body coordinate frame. The sideslip angle β and the side force F in the body coordinate frame are approximated as zero.

To simplify the analysis of the entry vehicle motion during an aeroassist maneuver, an equilibrium flight condition was considered in which the flight path angle (γ) is assumed to be sufficiently small and the rate of change of the flight path angle is zero. In addition, to simplify Equation (10), the gravitational force was assumed to be significantly

Table 5

Reference trajectory for explicit guidance.

V_{ref}	R_{ref}	D_{ref}	$\dot{h}_{\text{ref}}$	F_1	F_2	F_3
[km/s]	[km]	[G]	[m/s]	[km]	[km/G]	[km/(m/s)]
0.30	6.30	3.32	−166.33	22.59	−3.31	0.02
0.64	6.30	3.32	−166.33	40.41	−3.31	0.02
1.20	21.13	3.41	−207.58	80.15	−5.38	0.04
1.92	49.66	3.67	−175.15	179.09	−8.40	0.09
2.60	93.61	3.48	−122.98	347.14	−11.78	0.15
3.08	134.42	3.31	−151.73	522.63	−14.12	0.21
4.27	281.68	2.78	−151.73	610.05	−19.68	0.37
4.86	387.22	2.44	−188.47	862.11	−44.66	0.47
5.60	565.02	1.97	−193.64	1,264.36	−59.55	0.61
6.35	838.01	1.42	−239.36	1,815.89	−59.55	0.81
7.04	1,260.82	0.85	−285.08	2,565.02	−120.34	1.32
7.16	1,374.28	0.74	−330.80	2,792.82	−199.72	1.85
10.67	1,374.28	0.74	−376.52	2,792.82	−199.72	1.85

smaller than the drag force. Using the linearized equation of motion from the simplified Equation (10), the terminal condition was considered to occur when the range distance error to the target point was equivalent to zero. Equation (13) shows the command of the lift-to-drag ratio based on these assumptions:

$$\left(\frac{L}{D}\right)_{\text{command}} = \left(\frac{L}{D}\right)_{\text{current}} + F_1(R_{\text{current}} - R_{\text{guidance}}), \quad (13)$$

where R represents the range-to-go from the current position of the vehicle to its target landing point and F_1 is a variable gain. The current range-to-go, R_{current} , is calculated using the navigation outputs (ϕ_1, θ_1) and the geographic coordinates of the target landing point (ϕ_2, θ_2) with Equation (14).

$$R_{\text{current}} = \cos^{-1}\{\sin\phi_1\sin\phi_2 + \cos\phi_1\cos\phi_2\cos(\theta_2 - \theta_1)\}. \quad (14)$$

The value of R_{current} during each guidance cycle depends on the specific navigation policy used. The range-to-go is critical as it ensures that the command can appropriately adjust the lift-to-drag ratio for adequate guidance to the target point. Equation (15) expresses the relationship between the bank angle and lift-to-drag ratio:

$$\sigma_{\text{command}} = \cos^{-1} \frac{\left(\frac{L}{D}\right)_{\text{command}}}{\left(\frac{L}{D}\right)_{\text{vehicle}}}, \quad (15)$$

where the bank angle is expressed as the ratio of the vertical component of the lift-to-drag ratio to the maximum lift-to-drag ratio.

3.2. Explicit guidance method using reference trajectory

The range-to-go in the guidance cycle, R_{guidance} , is derived from the equation of motion of the vehicle in equilibrium flight, displayed in Equation (16), where R_{ref} is the approximate range-to-go value expressed as Equation (17) and D_{ref} represents the approximate drag acceleration expressed in Equation (18). The reference altitude rate, $\dot{h}_{\text{ref}}$, can be calculated from the reference velocity and lift-to-drag ratio using Equation (19), where r_E is the radius of the Earth and the values of D and $\dot{h}$ are measured during the navigation cycle. The gain parameters F_1, F_2 , and F_3 were calculated using the bias factors K_1, K_2 , and K_3 via Equations (20)–(22). The reference trajectories are listed in Table 5.

$$R_{\text{guidance}} = R_{\text{ref}} + F_2(D - D_{\text{ref}}) + F_3(\dot{h} - \dot{h}_{\text{ref}}), \quad (16)$$

$$R_{\text{ref}} = -\frac{1}{2} \left(\frac{L}{D}\right) \log\left(1 - \frac{V^2}{gr_E}\right), \quad (17)$$

$$D_{\text{ref}} = \frac{1}{\left(\frac{L}{D}\right)} \left(1 - \frac{V^2}{gr_E}\right), \quad (18)$$

$$\dot{h}_{\text{ref}} = \left(\frac{L}{D}\right)(V_0 - V), \quad (19)$$

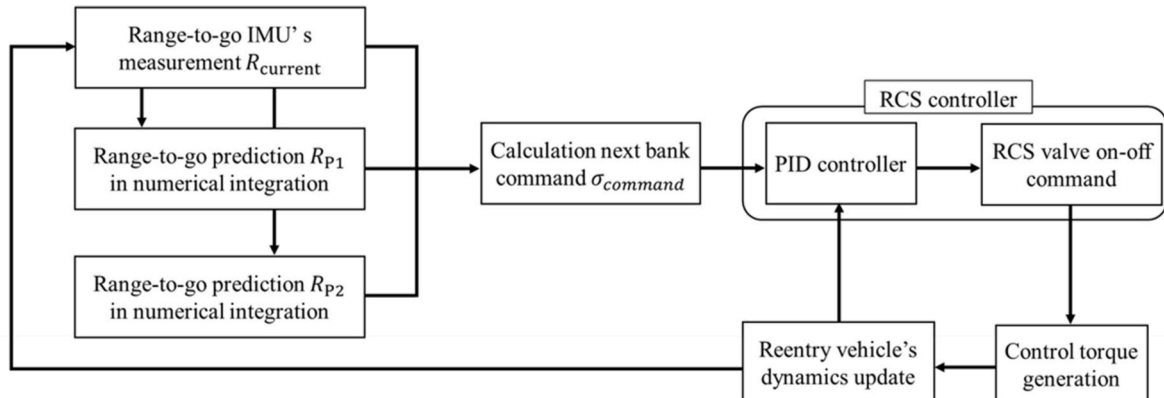
$$F_1 = \frac{4}{-K_1 \log\left(1 - \frac{V^2}{gr_E}\right)}, \quad (20)$$

$$F_2 = K_2 \frac{R_{\text{ref}}^2}{\left(\frac{L}{D}\right)}, \quad (21)$$

$$F_3 = K_3 \frac{R_{\text{ref}}^2}{\left(\frac{L}{D}\right)}. \quad (22)$$

3.3. Real-time predictor-corrector guidance

In this study, the range-to-go in the guidance cycle was applied to the predicted value using numerical integration, in which the range-to-go at the terminal point was predicted based on the current position of the vehicle. The variable gain is derived based on the sensitivity between the predicted range-to-go R_{predict1} at the bank angle, σ_{current} , and the predicted range-to-go R_{predict2} at the bank angle, $\sigma_{\text{current}} + \Delta\sigma$. Equation (23) expresses the bank angle for the next command:

**Fig. 10.** Guidance flow in the real-time predictor-corrector method.

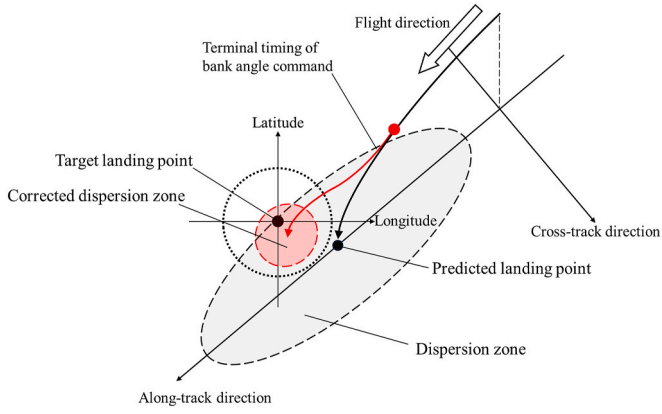


Fig. 11. Concept of terminal bank angle guidance.

bank angle command is determined using the secant method, as:

$$\sigma_{i+2} = \sigma_{i+1} - R_{\text{predict}}(\sigma_{i+1}) \frac{\sigma_{i+1} - \sigma_i}{R_{\text{predict}}(\sigma_{i+1}) - R_{\text{predict}}(\sigma_i)}, \quad (25)$$

where $R_{\text{predict}}(\sigma_i)$ is the prediction of the range-to-go on the ground to the target point. This iteration determines the desired bank angle command required to meet the landing error requirement by predicting the landing point of the capsule on the ground along a predicted trajectory at a constant bank angle. Fig. 11 shows a description of the terminal bank angle guidance. In this study, a landing error requirement of 1 km was applied for rapid convergence.

4. Simulation evaluation of aeroassist maneuvers

4.1. Simulation description

This section discusses the guidance performance of both the explicit and real-time predictor-corrector guidance methods, focusing on their

$$\cos\sigma_{\text{command}} = \cos\sigma_{\text{current}} + \frac{\cos(\sigma_{\text{current}} + \Delta\sigma) - \cos(\sigma_{\text{current}})}{R_{\text{predict}2} - R_{\text{predict}1}} (R_{\text{current}} - R_{\text{predict}1}). \quad (23)$$

Fig. 10 shows the guidance flow of the real-time predictor-corrector method. Equations of motion, non-dimensionalized by the radius of the Earth and orbital velocity around the Earth, were used for numerical integration in the guidance block.

The designed and actual values of the atmospheric density and aerodynamic characteristics of the capsule, such as the lift-to-drag ratio, may differ. Differences in the lift and drag forces affect the guidance performance significantly. As on-board estimations of the actual atmospheric density, lift coefficient, and drag coefficient during flight are challenging, the scale factors for the lift and drag forces were estimated using a first-order fading filter, as shown in Equation (24) [36].

$$\hat{X}^{n+1} = \beta \hat{X}^n + (1 - \beta) X^*, \quad (24)$$

where $\hat{X}^n$ is an estimated value of n-th number, β is a scale factor, and X^* is a ratio to the nominal value. The lift and drag forces in the body axis coordinate frame were measured using the acceleration and surface pressure of the capsule.

The cross-range guidance strategy involves the application of a bank angle reversal command. The cross-range deadband is related to the vehicle velocity, constant gain, and velocity bias. Here, gain and bias values were determined based on sensitivity analyses. However, this guidance, which relies on scheduling criteria, cannot provide precise guidance when uncertain errors occur. Conventional guidance approaches lack flexibility in terms of cross-range control. To overcome the limitations of aeroassist guidance, a terminal bank angle guidance method was developed in this study. The proposed method involves a lateral control approach that transitions from cross-range control based on the scheduled cross-range deadband to constant bank angle flight, which does not utilize the scheduled deadband. In a constant bank angle flight, the value of the

robustness against errors in atmospheric density and the lift-to-drag ratio. As mentioned in Section 2.4, the range-to-go at the entry interface differs by ~ 70 km owing to the disturbance moment caused by the HTS, specifically the offset of the center of gravity and the tilt angle of the HTS. Additionally, the distance is expected to increase further owing to other factors, such as navigation errors and the time delay of thruster ignition relative to the scheduled period. Similar to the process described in Section 2, the entry conditions at the entry interface were determined using the deorbit maneuver. The start time of the HTS firing was adjusted to verify multiple range-to-go distances at the entry interface, which are tabulated in Table 6. The total of 27 reentry cases, with range-to-go distances at the entry interface varying within 2,369–7,031 km at intervals of ~ 200 km, were used to select the range-to-go distance for a high guidance performance.

The shape of the entry capsule is similar to that of the HSRC model, with a lift-to-drag ratio of 0.20–0.30; therefore, the capsule was assumed to exhibit the aerodynamic characteristics of the HSRC model, showing similar axial and normal force coefficient values. Data from ground tests at Mach 0.4–10 were used for HSRC development. The lift and drag

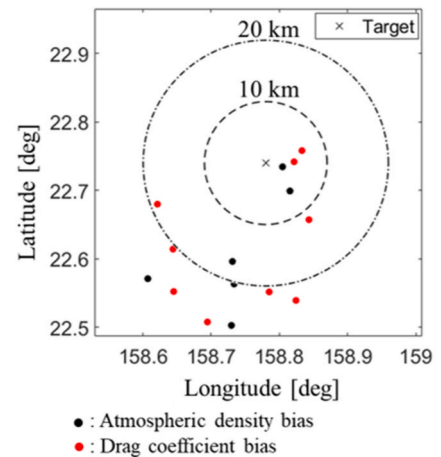


Fig. 12. Landing maps of explicit guidance with atmospheric density and drag coefficient bias.

Table 6

Deorbit maneuver conditions for range-to-go sensitivity analyses.

Property	Values
Total number of cases	27
Center of gravity [mm]	[30 20 20]
Elevation angle [°]	2.2
Deorbit ignition epoch	2026-07-05, 22:34:00–22:47:00, UTC
Deorbit ignition epoch interval [s]	30
Range-to-go in entry interface [km]	2,369–7,031

Table 7

Aeroassist maneuver conditions in explicit guidance.

Case	Range-to-go at entry interface [km]	Atmospheric density scale factor [%]	Drag coefficient scale factor [%]
1	3,824	100	100
2		90	100
3		110	100
4		100	90
5	4,011	100	110
6		100	100
7		90	100
8		110	100
9	4,190	100	90
10		100	110
11		100	100
12		90	100
13		110	100
14		100	90
15		100	110

coefficients were approximated for a blunt-nose entry capsule as a function of the angle of attack, as shown in Equation (26).

$$\begin{aligned} C_D &= C_A \cos \alpha + C_N \sin \alpha, \\ C_L &= -C_A \sin \alpha + C_N \cos \alpha. \end{aligned} \quad (26)$$

For Mach numbers beyond the ground test range, an upper limit of Mach 10 and a lower limit of Mach 0.4. were used. For conditions outside the experimental Mach number values, the axial and normal force coefficients were determined through interpolation.

After reentering the atmosphere, the vehicle flies in the Kepler orbit until the drag acceleration criterion is met. A negligible atmospheric density along with the constant bank angle of 0° and an angle of attack corresponding to the trim angle at Mach 10 were assumed during the Kepler orbit. Referencing Apollo entry-flight values, the drag acceleration criterion of 0.20 G was also assumed. On exceeding this threshold, the bank angle is determined by the guidance block. Here, the RCSs controlled at 50 Hz were used for the bank angle change. This control cycle referenced the flight values of the HSRC. Perfect navigation, i.e., no noise or bias in translational and rotational dynamics, was assumed in this analysis. A 4th-order Runge–Kutta integration at 100 Hz was performed for calculations, and analyses were conducted on the same platform as the deorbit maneuver analysis using the U.S. Standard Atmosphere 1976 model for atmospheric density. The gravity model uses a simplified sphere model without the harmonic term, while the elliptical gravity model with the harmonic term differs by only a few percent from the gravity model with a spherical shape. Here, a wind model of the upper atmosphere was not considered. Explicit guidance was used at 2 Hz, similar to the Apollo mission, whereas real-time predictor-corrector guidance was used at 1 Hz, similar to the HSRC mission. Moreover, like the conditions of the HSRC, guidance was continued until a Mach 3 vehicle velocity, complicating aerodynamic maneuvering in the upper atmosphere. Moreover, the analysis was terminated when the velocity became <240 m/s, which is the criterion for parachute deployment.

Uncertainty factors were used to ensure that atmospheric density and aerodynamic characteristics were not overlooked in calculations. Shape changes of the capsule owing to surface recession during reentry impact its aerodynamic properties significantly. To account for this effect, the measurement results from the HSRC ground experiment and calculation

Table 8

Flight performance in explicit guidance.

Case	Maximum aerodynamic load [G]	RCS total impulse [Ns]	Maximum atmospheric heating rate [MW/m ²]	Total atmospheric heating [MJ/m ²]
Mean	5.14	176.64	0.86	208.59
Variance (2-sigma)	1.12	77.65	0.044	17.34

results from the computer fluid dynamic analyses were compared; this comparison indicates that the lift-to-drag ratio may change by $\sim 10\%$ compared with the nominal shape. Additionally, the offset of the center of gravity may affect the aerodynamic performance of the capsule; however, comprehensively evaluating this effect is challenging. To manage variations in aerodynamic characteristics, the guidance algorithm must be significantly robust. Here, variations in atmospheric density and the lift-to-drag ratio were modeled using a uniform distribution as the scale factors.

4.2. Explicit guidance with atmospheric density and lift-to-drag ratio uncertainties

In this study, the performance of the explicit guidance algorithm, including its distribution, atmospheric density, and drag coefficient, was evaluated. With no bias in atmospheric density or drag coefficient, the range-to-go distance of 4,011 km was considered at the entry interface to ensure a minimum landing error relative to the target. Under these conditions, the cross- and down-range errors on the ground were -0.158 km and -4.322 km, respectively. Fig. 12 shows the landing dispersion across the 15 cases considered in this study with range-to-go values of 3,824, 4,011, and 4,190 km at the entry interface. For each distance, atmospheric density biases of -10% , nominal, and 10% , along with drag coefficient biases of -10% , nominal, and 10% , were considered, as listed in Table 7. Across the 15 cases, the mean cross-range and down-range errors are 12.68 and 8.17 km, respectively, with a landing point 2-sigma dispersion of 18.52 and 14.43 km for the cross-range and down-range, respectively. Thus, the explicit guidance method is not robust against a $\pm 10\%$ variation in atmospheric density and the lift-to-drag ratio. The reference trajectory is highly sensitive to changes in these parameters; although the reference trajectory is typically fine-tuned through trial and error, this process primarily relies on the empirical knowledge of the designer and does not guarantee robustness against uncertainties.

The total of the aerodynamic load, atmospheric heating rate, total atmospheric heating, and RCS total impulse for bank angle manipulation are crucial elements for evaluating the feasibility of a reentry flight. The aerodynamic load G can be calculated using Equation (27), and the atmospheric heat rate $\dot{q}$ can be expressed by Equation (28) based on the Detra–Kemp–Riddell theory [37].

$$G = \sqrt{D^2 + L^2}, \quad (27)$$

$$\dot{q} = 110.35 \sqrt{\frac{1}{R_n}} \sqrt{\frac{\rho}{\rho_0}} \left(\frac{V}{V_0} \right)^{3.15} \left(\frac{h_s - h_w}{h_s - h_{w0}} \right). \quad (28)$$

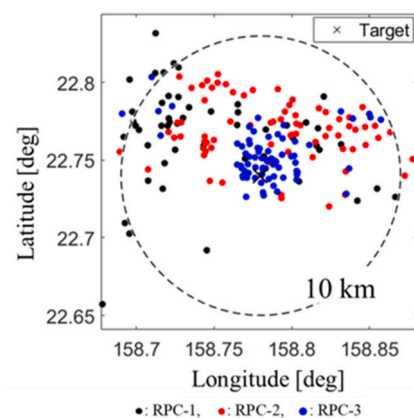


Fig. 13. Landing maps of the three types of real-time predictor-corrector guidance analyzed in this study with different atmospheric density and drag coefficient biases.

Table 9

Aeroassist maneuver conditions in real-time predictor-corrector guidance.

Parameter	Value
Range-to-go at entry interface [km]	5,462, 5,701, 5,886, 6,072, 6,258
Atmospheric density scale factor [%]	90, 100, 110
Drag coefficient scale factor [%]	80, 90, 100, 110, 120
Guidance approach [–]	RPC-1, RPC-2, RPC-3

Table 10

Cross-range errors in three types of real-time predictor-corrector guidance.

Range-to-go at entry interface [km]	RPC-1		RPC-2		RPC-3	
	Mean [km]	2-sigma [km]	Mean [km]	2-sigma [km]	Mean [km]	2-sigma [km]
5,462	–1.38	17.40	1.90	8.77	0.39	2.41
5,701	–4.35	32.05	1.61	9.45	0.55	6.29
5,886	–16.28	10.73	1.41	9.14	–0.85	5.60
6,072	–11.44	63.42	0.40	7.58	–0.10	5.69
6,258	–6.87	104.26	0.35	10.45	–0.28	7.51

Table 11

Down-range error in three types of real-time predictor-corrector guidance.

Range-to-go at entry interface [km]	RPC-1		RPC-2		RPC-3	
	Mean [km]	2-sigma [km]	Mean [km]	2-sigma [km]	Mean [km]	2-sigma [km]
5,462	2.39	13.30	0.79	8.74	0.53	2.62
5,701	–1.61	27.97	0.64	8.61	0.20	6.01
5,886	–1.14	54.66	0.40	10.17	1.62	7.01
6,072	7.71	53.71	–0.90	7.50	–0.42	4.56
6,258	–30.93	31.23	0.18	7.08	–0.39	5.37

The following parameters were assumed: R_n is the nose radius of the capsule, ρ_0 is the atmospheric density at sea level (1.225 kg/m^3), V_0 is the circular velocity in the LEO, h_s is the entropy of a stagnation point, h_w is the hot wall temperature ($1,111 \text{ K}$), and h_{w0} is the ground temperature (300 K). Here, the total atmospheric heating was calculated by summing the atmospheric heating rates during flight. Additionally, the total RCS impulse was determined by summing the thruster on time for roll-axis control. Table 8 summarizes the mean and variance of the maximum aerodynamic load, total RCS impulse, maximum atmospheric heating rate, and total heating. The results for aerodynamic load and the total RCS impulse indicate that explicit guidance significantly impacts

the conditions related to atmospheric density or aerodynamic characteristics. However, the atmospheric heating rate and total atmospheric heating showed minimal effects on model errors. In the HSRC case, the maximum allowable atmospheric heating rate was 1.00 MW/m^2 ; all 15 reentry cases meet this requirement. However, the total atmospheric heating of 200 MJ/m^2 was observed during shallow entry with the flight path angle of -1.1° , indicating that the total aerodynamic heating requirement may be particularly important for capsule safety.

4.3. Real-time predictor-corrector guidance with atmospheric density and lift-to-drag ratio uncertainties

The values listed in Table 6 were used as reentry conditions in this study. Uniform distributions of atmospheric density and drag coefficient were assumed with variations of $\pm 10\%$ and $\pm 20\%$ in intervals of 10% . As mentioned in Section 3.3, three types of real-time predictor-corrector guidance were implemented in this study: first, nominal guidance labeled RPC-1; second, a method that includes a lift-and-drag estimator labeled RPC-2; and third, a guidance that incorporates both the estimator and terminal bank angle guidance labeled RPC-3. Although there are errors, RPC-1 is incorrect for aerodynamic forces; lateral control in this method is based on the scheduled cross-range deadband. In contrast, RPC-2 can correct these forces from estimation. RPC-3 utilizes a similar correction method and enables autonomous lateral control without using the scheduled cross-range deadband to determine the terminal constant bank angle. Fig. 13 shows the landing dispersion in these three guidance approaches in 225 cases of range-to-go at an entry interface within 5,462–6,258 km, as listed in Table 9.

Fig. 13 shows that the landing dispersion can be gradually reduced for RPC-1, RPC-2, and RPC-3. Tables 10 and 11 summarize the results of the cross- and down-range errors, respectively, for 225 cases. In RPC-1, the range-to-go at an entry interface of $>5,701 \text{ km}$ cannot provide accurate guidance to the target point. In contrast, a significantly low landing dispersion is observed in RPC-2 with a lift-and-drag estimator, indicating that RPC-2 can precisely control the vehicle, even at long distances. Considering the mean and variance in RPC-2, the control accuracy remains consistent for range-to-go distances within 5,462–6,258 km. Additionally, the application of terminal bank angle guidance enhances the landing precision, resulting in a reduction in the landing dispersion for each range-to-go case. Notably, the control accuracy of RPC-3 is within 3 km of a range-to-go of 5,462 km, even in the presence of atmospheric density and drag coefficient errors, as listed in Table 9. Thus, both the estimator and terminal guidance functions effectively minimize the landing dispersion.

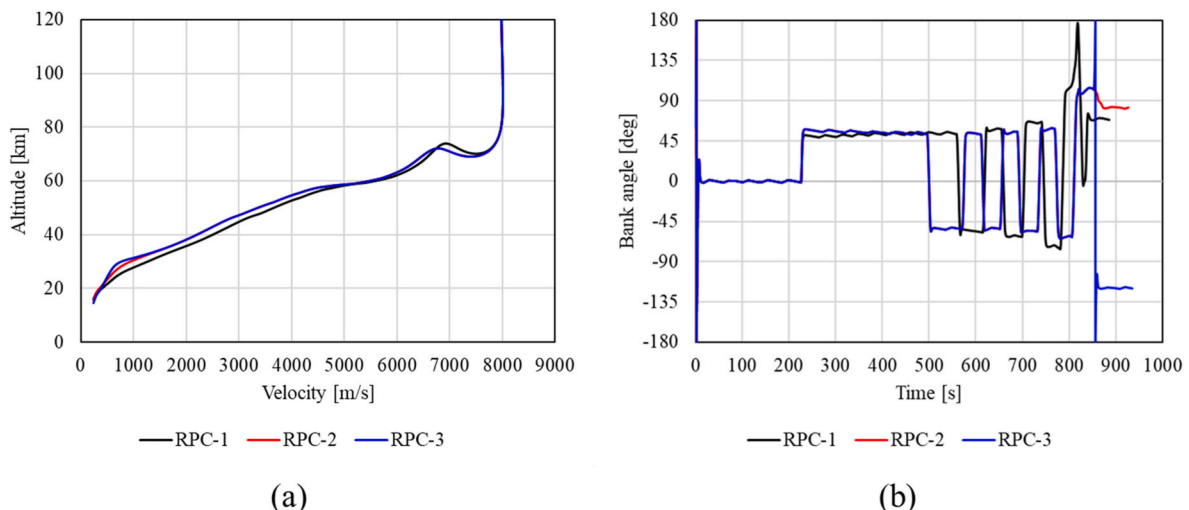


Fig. 14. Reentry profiles in three types of real-time predictor-corrector guidance: (a) Velocity versus altitude; (b) Bank angle profiles.

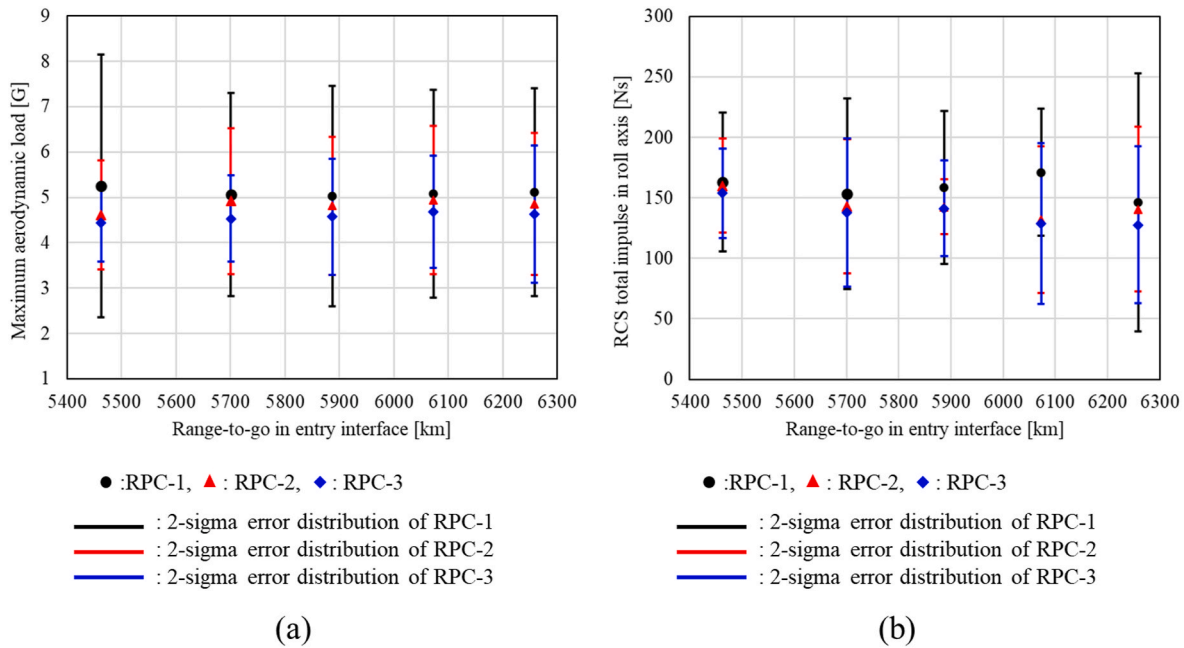


Fig. 15. Simulation results of the mean and variance values in three different real-time predictor-corrector approaches: (a) Aerodynamic load; (b) Total RCS impulse.

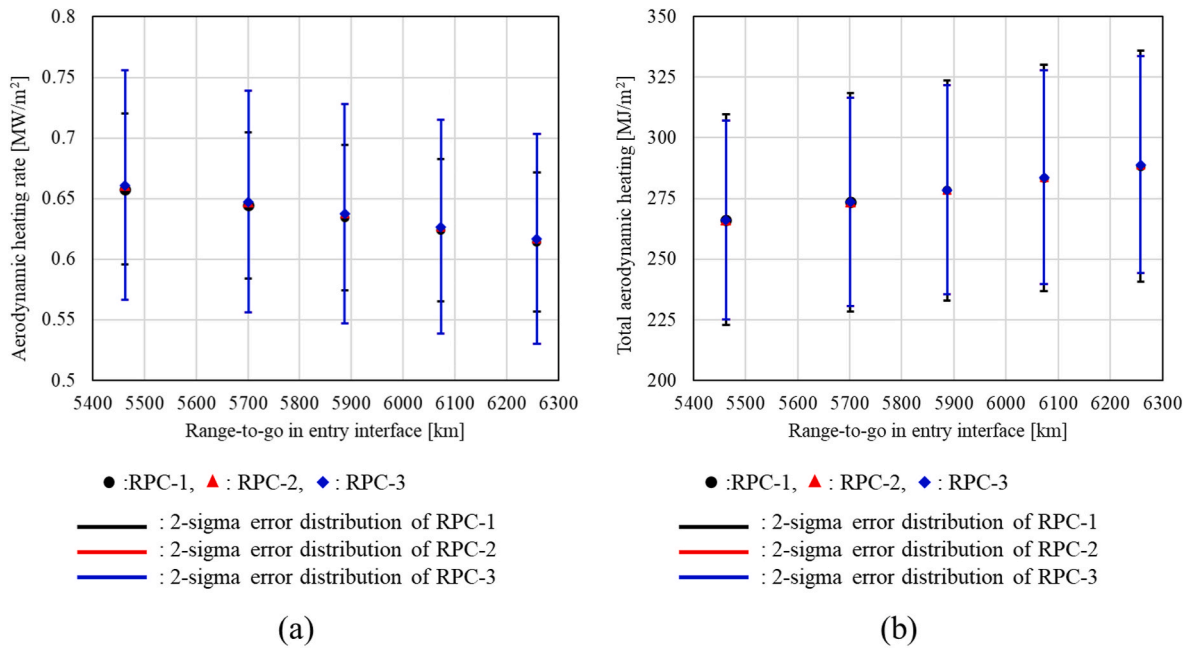


Fig. 16. Simulation results of the mean and variance values in three different real-time predictor-corrector approaches: (a) Aerodynamic heating rate, (b) Total aerodynamic heating. The error distribution of the RPC-2 is almost identical of the case of RPC-3.

Table 12				
Summary of real-time predictor-corrector guidance performances.				
	RPC-1	RPC-2	RPC-3	
Lift-and-drag estimator implementation	×	×	○	
Terminal bank angle guidance implementation	×	×	○	
Control accuracy	△	○	◎	
Maximum aerodynamic load	△	○	○	
RCS total impulse	△	○	○	

Fig. 14 shows the reentry profile of the altitude, velocity, and bank angle for the range-to-go distance of 5462 km. Fig. 14(a) indicates that the velocity profiles of RPC-2 and RPC-3 differ from those of RPC-1, and that the profiles of RPC-2 and RPC-3 are also slightly different at the altitude of 30 km. Fig. 14(b) shows the flown bank angle profile, highlighting the differences in the terminal bank angles among the three guidance cases. In the terminal guidance approach, the bank angle is determined to meet the landing error criteria, and bank reversals are executed, regardless of the scheduled cross-range deadband.

Next, the guidance performance, aerodynamic load, total RCS impulse for bank angle manipulation, atmospheric heating rate, and total heating are discussed. Fig. 15(a) shows the maximum aerodynamic load

for the three guidance methods analyzed in this study, along with the error distribution at 2-sigma. The maximum values are nearly the same across the three guidance approaches; however, the error distribution decreases on the estimator and terminal guidance application. Fig. 15(b) shows the total impulse of the RCS along the roll axis. Utilizing the estimator reduces propellant consumption owing to a reduction in the number of bank reversals, which improves lateral control. Fig. 16(a) and (b) show that the atmospheric heating rate and total atmospheric heating for RPC-1 are equivalent to those of RPC-2 and -3.

The guidance performances of the real-time predictor-corrector methods analyzed in this study are presented in Table 12. In the second and third rows, the symbols $\times$ and $\circ$ indicate whether the algorithm is implemented in each method. In the fourth through sixth rows, the symbols $\triangle$, $\circ$, and $\odot$ represent the performance of these methods. This study highlights the improvement in numerical integration guidance achieved using both the lift-and-drag estimator and terminal bank angle guidance. These techniques ensure the control accuracy of 3 km for the landing error, even with 10% error in atmospheric density and a 20% error in atmospheric characteristics. Thus, the proposed guidance method is highly robust against uncertainties in the entry trajectory, atmospheric density model, and the atmospheric characteristics of the capsule.

5. Conclusions

This study reports guidance techniques for the controlled Earth reentry of a small spacecraft that integrate an attitude control method during the deorbit maneuver with a hybrid thruster system and a guidance method during the aeroassist maneuver.

First, this paper describes the deorbit maneuver for Earth reentry using a hybrid thruster system developed to facilitate the deorbit maneuver for a small spacecraft. This thruster system enables a small spacecraft to exhibit a velocity increment of 100 m/s over 200 s of firing duration. During the thruster operation, attitude control is required to counteract the disturbance moment generated by the thrust force. Here, a spin stabilization approach was used as the attitude control method, considering the allowable disturbances and propellant consumption of the cold-gas thrusters in the system. Relevant uncertainties, such as the offset of the center of gravity of the spacecraft and the tilt angle of the hybrid thruster during the deorbit maneuver, were considered while evaluating the control performance. The proposed method facilitated the deorbit maneuver and could insert the spacecraft into a reentry trajectory with attitude disturbances, which caused the range dispersion of 70 km in the along-track direction and a negligible range dispersion in the cross-track direction.

Subsequently, a strategy for the aeroassist maneuver of a 150 kg small entry capsule that separates from the spacecraft after the deorbit maneuver was analyzed. The classical approach for aeroassist maneuvers uses explicit guidance based on a reference trajectory. However, as this approach depends on the scheduled flight profile, it cannot be applied under reentry conditions where errors may occur owing to orbital parameters at the entry interface, inaccuracies from the deorbit maneuver, and the atmospheric density and atmospheric force models of the capsule. In contrast, real-time predictor-corrector guidance for predicting the onboard trajectory is a numerical, non-scheduling approach. A highly accurate guidance system based on a real-time predictor-corrector method using a lift-and-drag onboard estimator and terminal bank angle guidance was developed in this study. The guidance performance of the proposed system was evaluated in terms of control accuracy and robustness against model errors in atmospheric density and characteristics. The developed guidance method shows a high control accuracy, exhibiting a dispersion of within 3 km in 2-sigma, even with 10% and 20% errors in the atmospheric density model and atmospheric force model of the capsule, respectively. The variation in the terminal landing point can be reduced to within several kilometers, even if the dispersion of the range at an entry interface reaches approximately 200

km. In addition, the proposed guidance approach exhibits beneficial aspects in terms of the maximum aerodynamic load and total RCS impulse.

In summary, the findings of this study are expected to contribute toward the development of a novel space transportation system for small spacecraft. Notably, besides Earth reentry, the deorbit control system can be used for Moon landings and Mars entry. The proposed aeroassist guidance system is expected to enable precise landings in Mars entry, potentially exhibiting a higher guidance accuracy than conventional explicit guidance systems. Furthermore, as a potential application, this technique can also be applied to larger spacecraft by mounting one or more hybrid thrusters. Regardless of the spacecraft's size, these strategies can contribute to active debris removal after completing orbital operations.

CRediT authorship contribution statement

Takumi Saito: Writing – original draft. **Toshinori Kuwahara:** Writing – review & editing. **Yuji Saito:** Writing – review & editing. **Yuji Sato:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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